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EXPERIMENT - 6

Snort IDS/IPS Using TryHackMe

Step 1: Login to Platform

Open TryHackMe in a browser and log in using your account.

Step 2: Open Snort Room

Search for a Snort-related room and join it to begin the experiment.

Step 3: Start AttackBox

Launch the AttackBox, which provides a ready-to-use Linux environment for performing the tasks.

Step 4: Verify Snort Installation

Open the terminal and confirm that Snort is installed and ready to use.

Step 5: Run Snort in IDS Mode

Start Snort in detection mode so that it monitors network traffic and generates alerts for suspicious activity.

Step 6: Detect Network Threats

Perform the tasks given in the room and observe how Snort identifies and reports potential threats in real time.

Step 7: Configure IPS Mode

Enable prevention mode to allow Snort to block malicious traffic based on predefined rules.

Step 8: Analyze Alerts

Review the alerts generated by Snort and identify different types of network attacks.

Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

170 ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 ✓ Correct Answer

Keep reading the output.How many "HTTP response headers" were extracted?

3 ✓ Correct Answer

Investigate the **mx-1.pcap** file **with the second** configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

68 ✓ Correct Answer

Investigate the **mx-2.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-2.pcap
```

What is the number of the generated alerts?

340 ✓ Correct Answer

Keep reading the output. What is the number of the detected TCP packets?

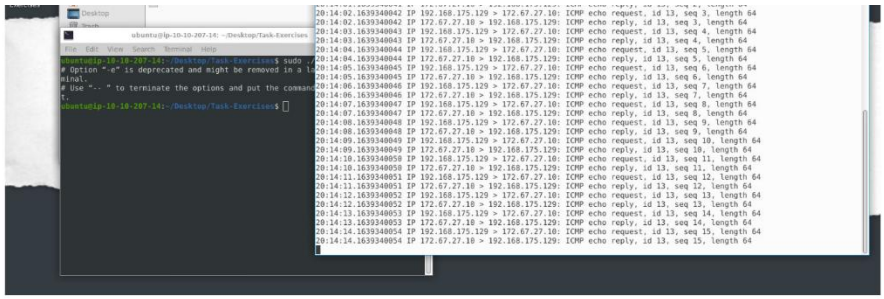
82 ✓ Correct Answer

Investigate the **mx-2.pcap and mx-3.pcap** files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"
```

What is the number of the generated alerts?

1020 ✓ Correct Answer



Answer the questions below

Navigate to the Task-Exercises folder and run the command "./.easy.sh" and write the output

Too Easy! ✓ Correct Answer

Task 3 ✓ Introduction to IDS/IPS

-V / --version	This parameter provides information about your instance version.
-c	Identifying the configuration file
-T	Snort's self-test parameter allows you to test your setup with this parameter.
- q	Quiet mode prevents Snort from displaying the default banner and initial information about your setup.

That was easy; let's move on to the next task and continue exploring snort modes.

Answer the questions below

Run the Snort instance and check the build number.

149 ✓ Correct Answer

Test the current instance with "/etc/snort/snort.conf" file and check how many rules are loaded with the current build.

4151 ✓ Correct Answer

Test the current instance with "/etc/snort/snortv2.conf" file and check how many rules are loaded with the current build.

1 ✓ Correct Answer

```
12/18-07:40:02.550844  [Drop] [**] [1:1000001:0] ICMP Packet found [**] [Priority: 0] {ICMP} 172.217.169.142 -> 192.168.1.18
^C*** Caught Int-Signal
```

As you can see in the picture above, Snort blocked the packets this time. **We used the same rule with a different action (drop/reject).** Remember, for the scope of this task, our point is the operating mode, not the rule.

Answer the questions below

Investigate the traffic with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l .
```

Execute the traffic generator script and choose **"TASK-7 Exercise"**. Wait until the traffic stops, then stop the Snort instance. Now analyse the output summary and answer the question.

```
sudo ./traffic-generator.sh
```

What is the number of the detected HTTP GET methods?

2 ✓ Correct Answer

You can practice the rest of the parameters by using the traffic-generator script.

No answer needed ✓ Correct Answer

Investigate the **mx-1.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

170 — ✓ Correct Answer

Keep reading the output. How many TCP Segments are Queued?

18 — ✓ Correct Answer

Keep reading the output.How many "HTTP response headers" were extracted?

3 — ✓ Correct Answer

Investigate the **mx-1.pcap** file **with the second** configuration file.

```
sudo snort -c /etc/snort/snortv2.conf -A full -l . -r mx-1.pcap
```

What is the number of the generated alerts?

68 — ✓ Correct Answer

Investigate the **mx-2.pcap** file with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . -r mx-2.pcap
```

What is the number of the generated alerts?

340 — ✓ Correct Answer

Keep reading the output. What is the number of the detected TCP packets?

82 — ✓ Correct Answer

Investigate the **mx-2.pcap and mx-3.pcap** files with the default configuration file.

```
sudo snort -c /etc/snort/snort.conf -A full -l . --pcap-list="mx-2.pcap mx-3.pcap"
```

What is the number of the generated alerts?

1020 — ✓ Correct Answer

Result

Snort was successfully used as an IDS/IPS to detect and prevent network threats, demonstrating real-time network security monitoring in a controlled environment.